Lecture 10: LTI Resistive Circuits and Equivalent One-Ports

EE201 – Circuit Theory I

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current voltage a part of a circuit (box N)

Review: equations from the graph alone

	incidence / mesh	**fundamental cutset / loop**
KCL	$A\mathbf{i} = \mathbf{0}$	$Q\mathbf{i} = \mathbf{0}$
KVL	$M\mathbf{v} = \mathbf{0}$	$B\mathbf{v} = \mathbf{0}$
$n-1$ voltages	node voltages $\mathbf{e}$	tree voltages $\mathbf{v}_T$
$b-n+1$ currents	mesh currents $\mathbf{i}_M$	co-tree currents $\mathbf{i}_C$
works for	M: planar, nonseparable	any connected graph

KCL and KVL depend only on the graph: they hold for any elements.

A circuit has $2b$ unknowns: b more equations come from the elements.

Why this lecture?

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  - substitution theorem, equivalent one-ports
  - series and parallel connections, source transformation

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- **This lecture — transform:** replace a part by a source, or by a simpler part with the same terminal behaviour
  - substitution theorem, equivalent one-ports
  - series and parallel connections, source transformation
- **Later — the circuit as it is:** node voltage and mesh current analysis

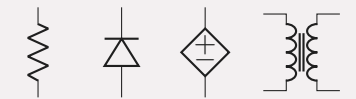
Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain
2. **A first example** — simplify a circuit step by step
3. **Substitution theorem** — replace a part of a circuit by a source
4. **One-ports and equivalence** — the same i–v characteristic at the port
5. **Series and parallel** — combine resistors and sources
6. **One source and one resistor** — exchange a voltage source and a current source

Resistive and dynamic elements

Resistive element: no memory

Its law relates v and i at the same instant, algebraically.

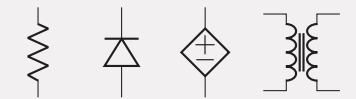


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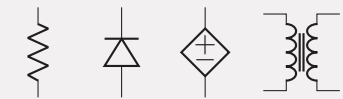


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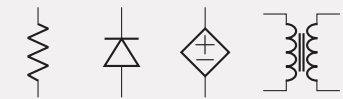
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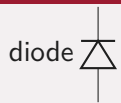
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"Resistive" means memoryless, not only "a resistor".

An element is linear if its law is linear in v, i; time-invariant if its law does not change with time.

Three questions about a circuit

Linear?

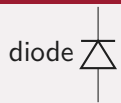


Nonlinear if it contains at least one nonlinear element, e.g. a diode.

Otherwise **linear**.

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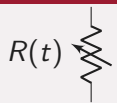
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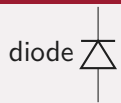


Time-varying if it contains at least one time-varying element, e.g. $R(t)$.

Otherwise **time-invariant**.

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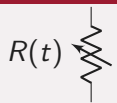
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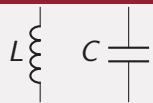
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Resistive?

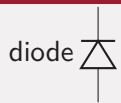


Dynamic if it contains at least one dynamic element: an L or a C.

Otherwise **resistive**.

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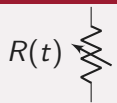
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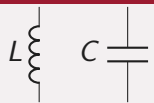
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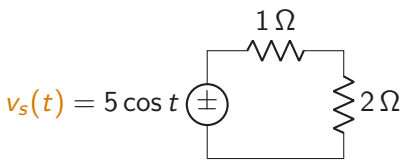
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LTI resistive circuit = linear + time-invariant + resistive:

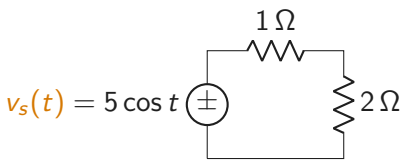
apart from the independent sources, every element is linear, time-invariant and resistive.

Independent sources are inputs



The source varies in time. Is the circuit time-varying?

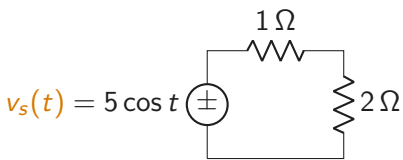
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No. An independent source is an external input: a given signal of time.

The three questions are about the circuit that the inputs drive, not about the inputs. This is an LTI resistive circuit.

What varies in time?

In every circuit $v(t)$ and $i(t)$ are functions of time. The question is whether the element law changes with time.

Time-invariant resistor

$$v(t) = R\, i(t)$$

the same rule at every instant

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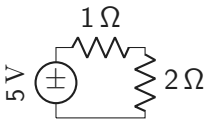
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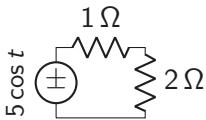
Linear or nonlinear is also a property of the element law: $v = Ri$ is linear, a diode law is not.

Which circuit is which?

Same circuit, two inputs

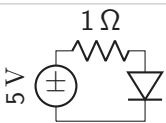


(a)

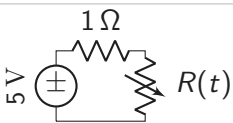


(b)

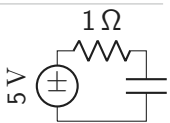
Same input, 2 Ω replaced



(c) diode



(d)



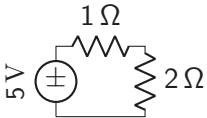
(e)

A source, a 1 Ω resistor and one more element. In (d), $R(t) = 2 + \sin t\ \Omega$.

Linear? Time-invariant? Resistive?

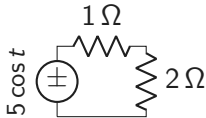
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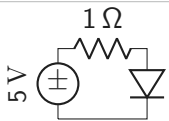
LTI resistive



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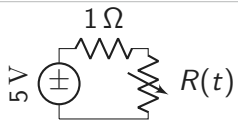
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Same input, 2 Ω replaced



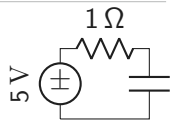
(c) diode

nonlinear



(d)

time-varying



(e)

dynamic

The input does not change the class: (a) and (b) are both LTI resistive.

Each of (c), (d), (e) breaks exactly one of the three properties.

What we gain

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$$v = Ri \quad \text{(resistor)}, \qquad v = \mu v_x, \quad i = g v_x, \quad \ldots \quad \text{(dependent sources)}$$

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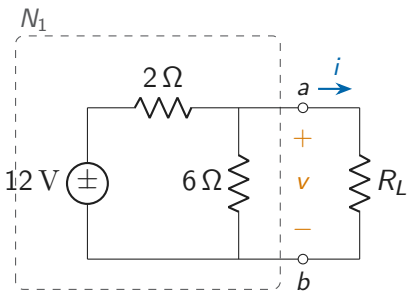
- Time enters only through the values of the independent sources.
- So at each instant the circuit is solved like a circuit with constant sources: $2b$ linear algebraic equations.

Roadmap

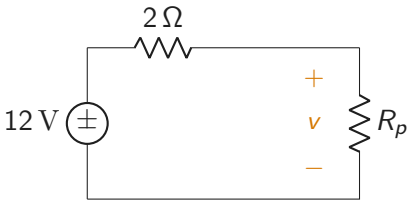
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A first example: combine the resistors

R_L (the load) can be any resistive element.
Find v and i.



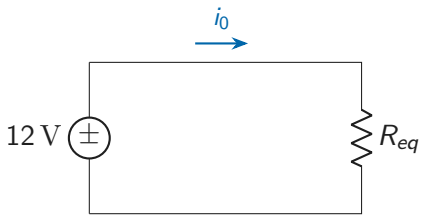
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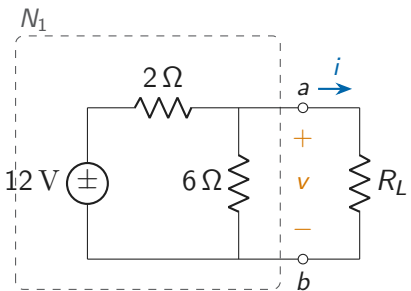
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$$R_{eq} = 2 + R_p = \frac{12 + 8R_L}{6 + R_L}, \qquad i_0 = \frac{12}{R_{eq}}$$

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Step 3. Back to the circuit:

$$v = R_p\, i_0 = \frac{72R_L}{12+8R_L}$$

$$i = \frac{v}{R_L} = \frac{72}{12+8R_L}$$

Read the answer as a circuit

$$i = \frac{72}{12 + 8R_L} = \frac{9}{1.5 + R_L}$$

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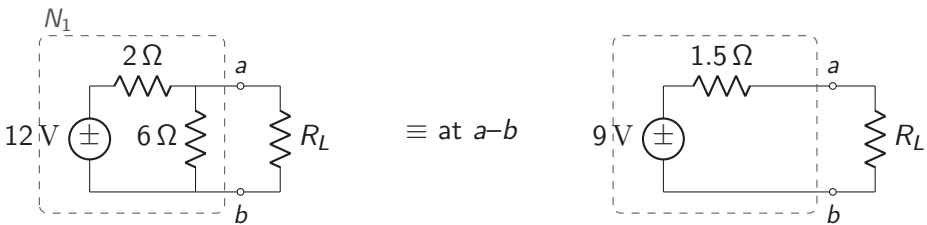
$$i = \frac{72}{12 + 8R_L} = \frac{9}{1.5 + R_L}$$

- The denominator is a constant, $1.5\,\Omega$, plus R_L.
- A $9\,\mathrm{V}$ source driving $1.5\,\Omega$ and R_L in series gives the same i.

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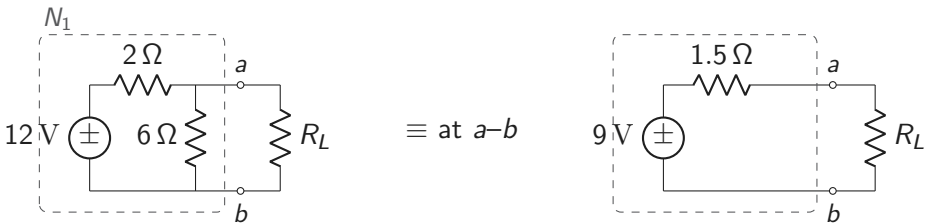
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The part left of a–b is now one source and one resistor.

For $R_L = 3\,\Omega$: $i = 2\,\mathrm{A}$, $v = 6\,\mathrm{V}$.

What did we just do?

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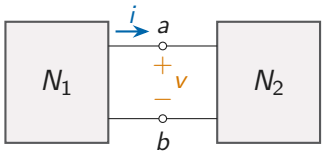
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question	**answered by**
May a part be replaced by a source?	substitution theorem
When may a part be replaced by a simpler part?	equivalent one-ports
Which simple replacements are there?	series, parallel, source transformation

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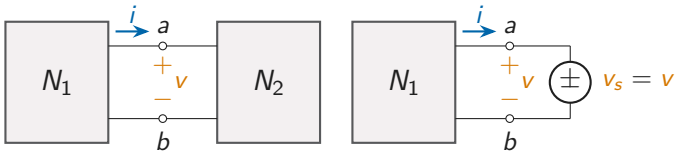
The substitution theorem



(I)

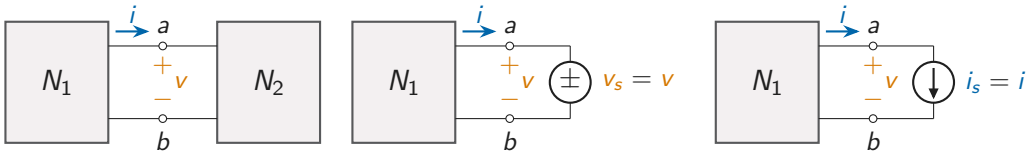
A circuit cut into two parts, joined at the terminals a, b.
i: current from N_1 into N_2 v: voltage between a and b

The substitution theorem



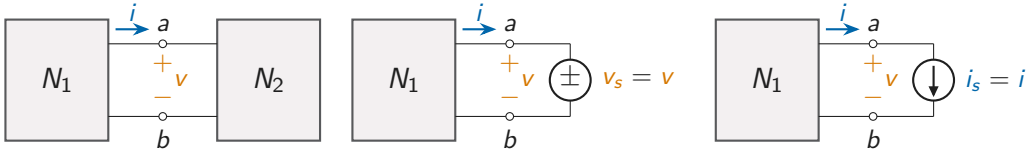
Replace N_2 by a voltage source with $v_s(t) = v(t)$ ...

The substitution theorem



... or by a current source with $i_s(t) = i(t)$.

The substitution theorem



Substitution theorem

No branch voltage or branch current inside N_1 changes. An observer inside N_1 cannot tell (I), (II) and (III) apart.

Why does it work?

- The equations inside N_1 (KCL, KVL, element equations) do not change.

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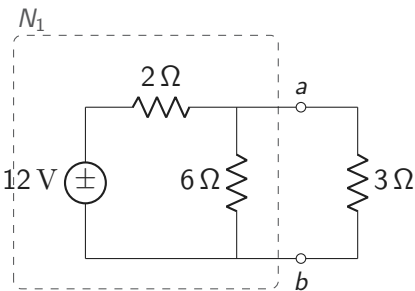
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One condition

The new circuit must have a unique solution.

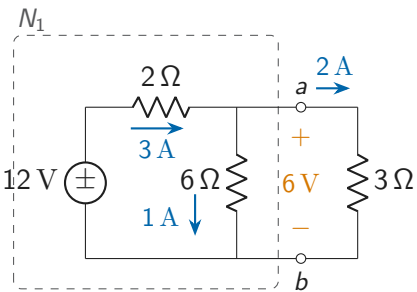
Example: N_1 a single current source and N_2 replaced by a current source $\Rightarrow$ v undetermined.

Example: substitution



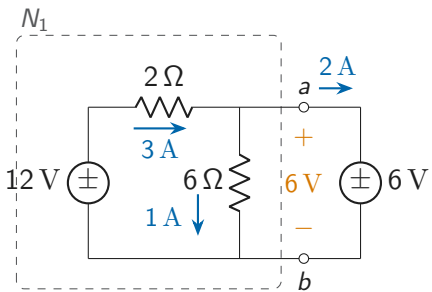
The first example with $R_L = 3\,\Omega$, cut at a, b: N_2 is the load.

Example: substitution



(I) From the first example ($R_L = 3\,\Omega$): $v = 6\,\mathrm{V}$, $i = 2\,\mathrm{A}$; inside N_1: $3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$.

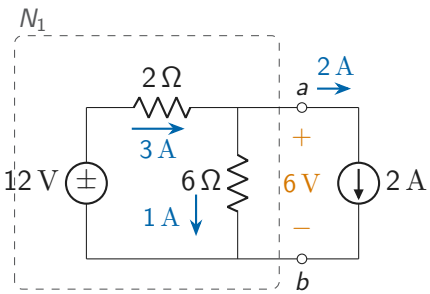
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(III) A $2\,\mathrm{A}$ source: KCL at a, $\frac{12-v}{2} = \frac{v}{6} + 2$, so $v = 6\,\mathrm{V}$: again $3\,\mathrm{A}$ and $1\,\mathrm{A}$. ✓

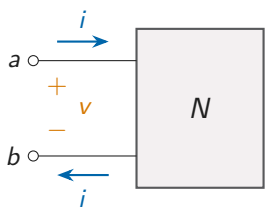
Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain ✓
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Ports and one-ports

Port, one-port

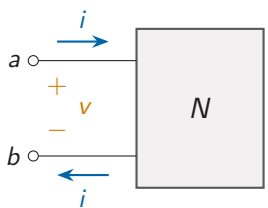
A pair of terminals is a port if the current entering one terminal equals the current leaving the other. A one-port is a circuit, or a circuit element, with one pair of terminals accessible from outside.



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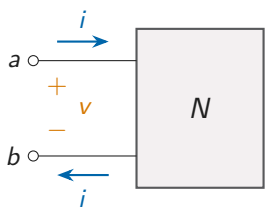


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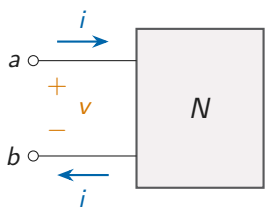


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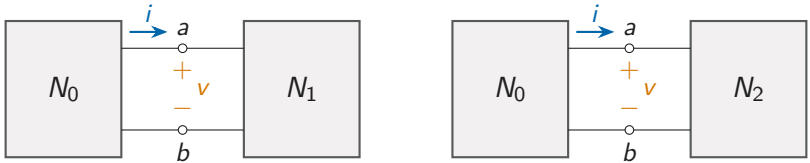


- A one-port is always a port: KCL on a closed surface around N.
- i enters the $+$ terminal of v.
- The relation between v and i is the i–v characteristic, e.g. $v = Ri$ for a resistor.

Equivalent one-ports

Equivalent one-ports

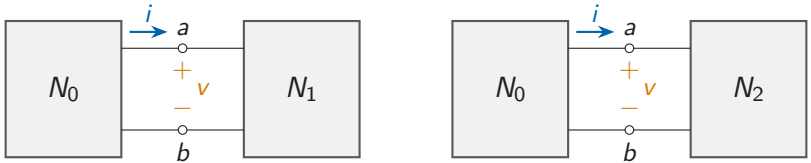
$N_1 \equiv N_2$ if they have the same i–v characteristic at their terminals.



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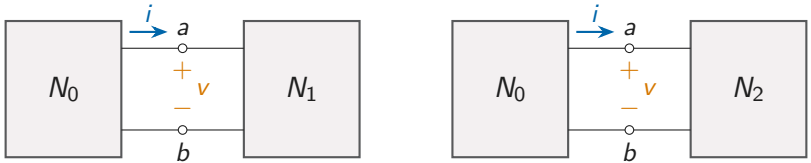


For any N_0: every branch voltage and current inside N_0 is the same in both.

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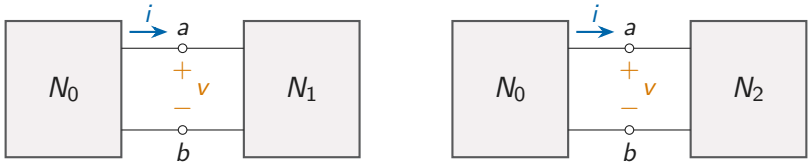
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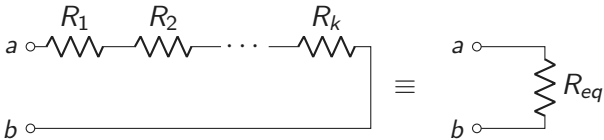
Inside N_1 and N_2 the branch values may differ.

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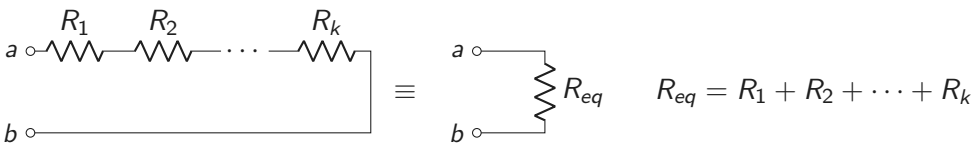
Series connection

Same current i through every element (KCL); the voltages add (KVL).



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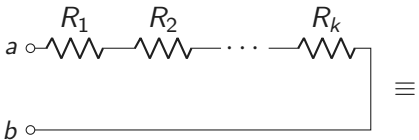


$$R_{eq} = R_1 + R_2 + \cdots + R_k$$

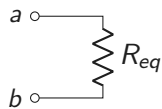
$$v = R_1 i + R_2 i + \cdots + R_k i = R_{eq} i$$

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$\equiv$

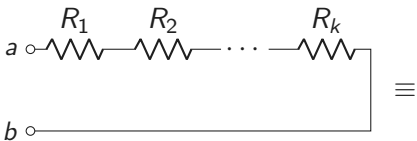


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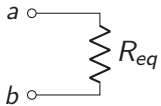
Voltage divider: the share of R_j is $v_j = \dfrac{R_j}{R_{eq}}\, v$

Series connection

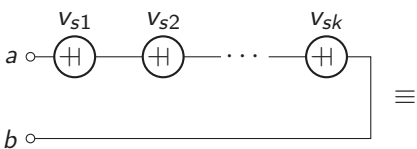
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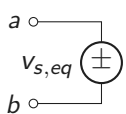
$\equiv$



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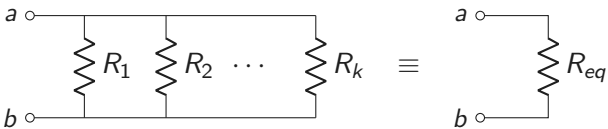


$$v_{s,eq} = v_{s1} + v_{s2} + \cdots + v_{sk}$$

All $+$ signs towards a; a source turned the other way enters with $-$.

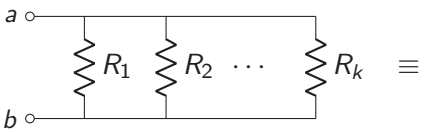
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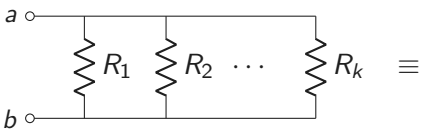


$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_k}$$

With $G_j = 1/R_j$: $\quad i = G_1 v + \cdots + G_k v = G_{eq} v, \quad G_{eq} = G_1 + \cdots + G_k$

Parallel connection

Same voltage v across every element (KVL); the currents add (KCL).

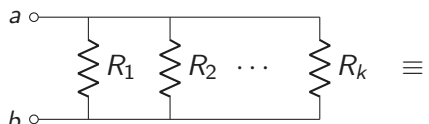


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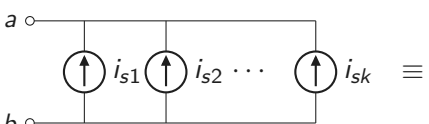
Current divider: $i_j = \dfrac{G_j}{G_{eq}}\, i.$ Two resistors: $R_{eq} = \dfrac{R_1 R_2}{R_1 + R_2}$

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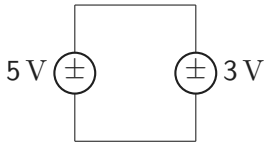
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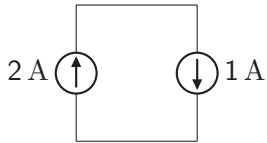
$$i_{s,eq} = i_{s1} + i_{s2} + \cdots + i_{sk}$$

All arrows towards a; a source turned the other way enters with $-$.

Two connections to avoid



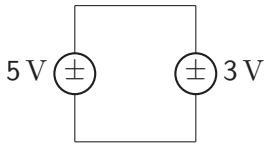
two voltage sources in parallel



two current sources in series

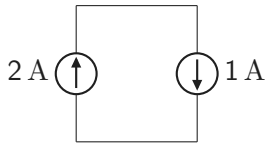
Are these allowed?

Two connections to avoid



two voltage sources in parallel

KVL fails: $5 \neq 3$.



two current sources in series

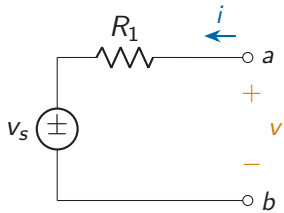
KCL fails: $2 \neq 1$.

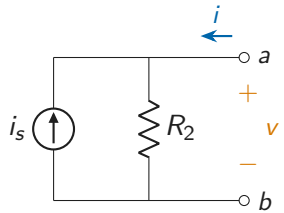
With equal values they are allowed, but the shared current (or voltage) is not fixed.

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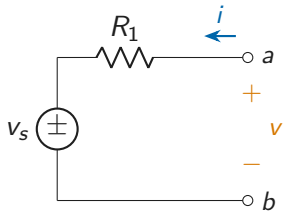
One source and one resistor

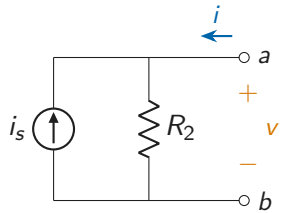




When do these two one-ports have the same i–v characteristic?

One source and one resistor

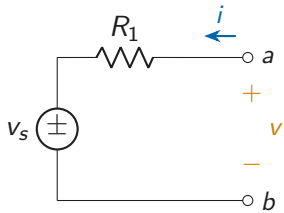




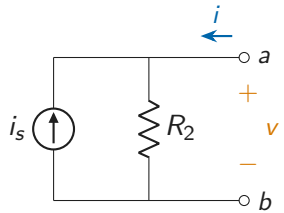
$$v = R_1 i + v_s$$

KVL around the loop: R_1 carries i.

One source and one resistor



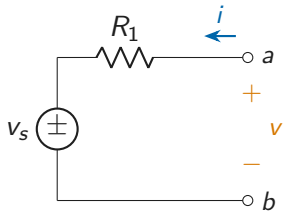
$$v = R_1 i + v_s$$



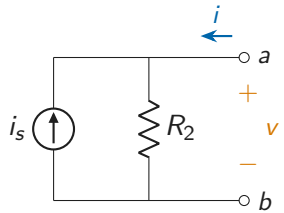
$$v = R_2(i + i_s) = R_2 i + R_2 i_s$$

i and i_s both flow into the top node: R_2 carries $i + i_s$.

One source and one resistor



$$v = R_1 i + v_s$$



$$v = R_2(i + i_s) = R_2 i + R_2 i_s$$

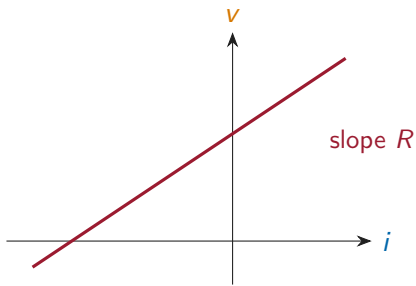
Both are straight lines in the i–v plane: equal slopes and equal intercepts.

Source transformation

Source transformation

The two one-ports are equivalent if

$$R_1 = R_2 = R \qquad \text{and} \qquad v_s = R\, i_s.$$



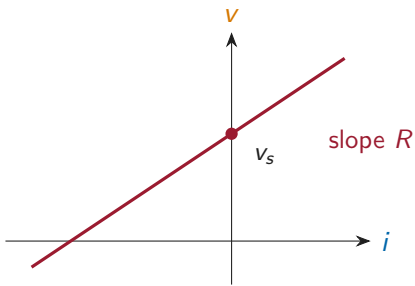
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Open circuit ($i = 0$): $v = v_s$.



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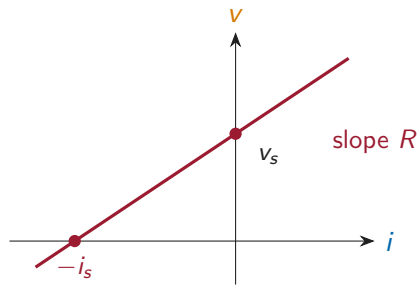
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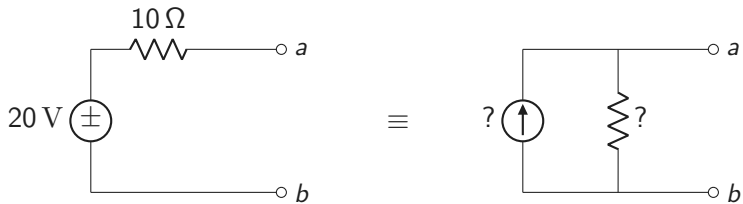
$$R_1 = R_2 = R \qquad \text{and} \qquad v_s = R\, i_s.$$

Open circuit ($i = 0$): $v = v_s$.

Short circuit ($v = 0$): $i = -i_s$.

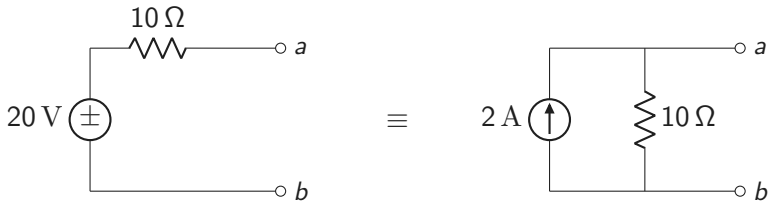


Example: a source transformation



Find the current source and the resistor.

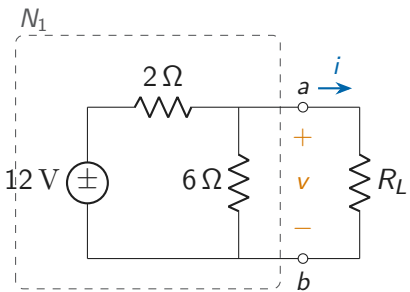
Example: a source transformation



$$R = 10\,\Omega, \qquad i_s = v_s/R = 20/10 = 2\,\mathrm{A}$$

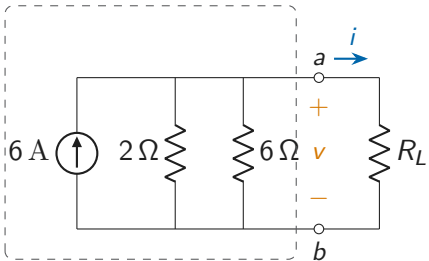
Both: open-circuit voltage $20\,\mathrm{V}$, short-circuit current $2\,\mathrm{A}$ (out of a).

The first example, again



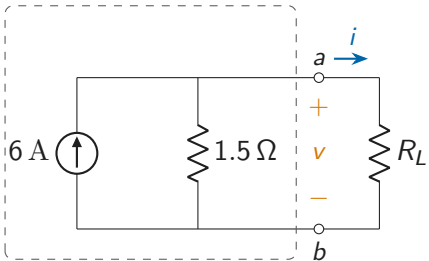
Now simplify only the part left of a–b; R_L stays.

The first example, again



Step 1. Source transformation:
$12\,\mathrm{V}$ with $2\,\Omega \rightarrow 6\,\mathrm{A}$ with $2\,\Omega$ $(12 = 2 \times 6)$

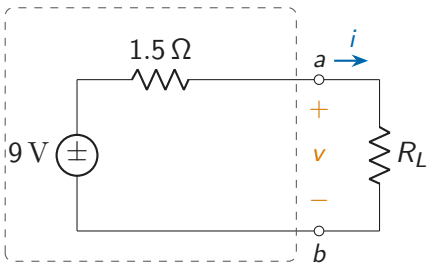
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$12\,\mathrm{V}$ with $2\,\Omega \rightarrow 6\,\mathrm{A}$ with $2\,\Omega$ $(12 = 2 \times 6)$

Step 2. Parallel: $2\,\Omega \parallel 6\,\Omega = 1.5\,\Omega$

The first example, again

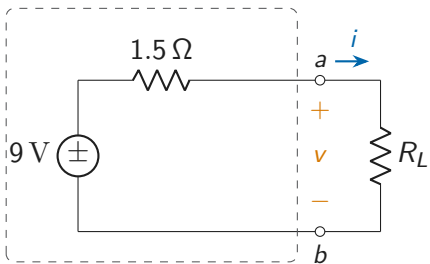


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$6\,\mathrm{A}$ with $1.5\,\Omega \rightarrow 9\,\mathrm{V}$ with $1.5\,\Omega$
$(9 = 1.5 \times 6)$

The first example, again



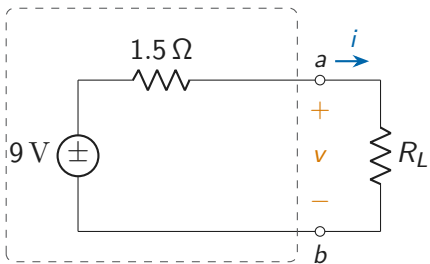
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Same result, and R_L never enters the algebra.

The first example, again



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$(9 = 1.5 \times 6)$

Same result, and R_L never enters the algebra.

Careful: inside, the values differ; for $R_L = 3\,\Omega$ the $12\,\mathrm{V}$ source carries $3\,\mathrm{A}$, the $9\,\mathrm{V}$ source $2\,\mathrm{A}$.

Solving LTI resistive circuits

- Series and parallel connections and source transformations simplify many circuits, but not all.

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  1. node voltage analysis: the $n-1$ node voltages are the unknowns;
  2. mesh current analysis: the $b-n+1$ mesh currents are the unknowns.
- Both start from the matrices met earlier (A, M), now with linear element equations.

Summary

- LTI resistive: apart from the independent sources (external inputs), every element is linear, time-invariant and memoryless, so all equations are linear and algebraic.
- Substitution: replace N_2 by a voltage source $v_s = v$ or a current source $i_s = i$; nothing changes inside N_1.
- Equivalent one-ports: the same i–v characteristic; the rest cannot tell them apart.
- Series: $R_{eq} = \sum R_j$, $v_{s,eq} = \sum v_{sj}$. Parallel: $G_{eq} = \sum G_j$, $i_{s,eq} = \sum i_{sj}$.
- Source transformation: v_s with series $R \equiv i_s = v_s/R$ with parallel R.